

REAL-TIME IMAGING FOR COVID-19 & BREAST CANCER DETECTION

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**REAL-TIME IMAGING FOR COVID-19 &
BREAST CANCER DETECTION**

A PROJECT REPORT

**submitted in partial fulfillment of the
requirements for the award of**

BACHELOR OF TECHNOLOGY

in

ELECTRONICS & COMMUNICATION ENGINEERING

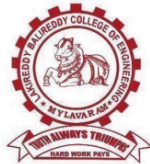
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Affiliated to JNTUK, Kakinada & Approved by AICTE, New Delhi

Accredited by NBA (under tier-I) and NAAC

Certified by ISO 21001:2018

2025- 2026

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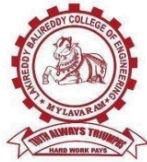
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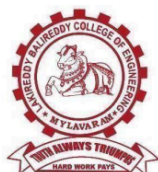
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CERTIFICATE

This is to certify that project work entitled “**REAL-TIME IMAGING FOR COVID-19 & BREAST CANCER DETECTION**” is a bonafide work done and submitted by **P.KALYANI (22761A04H9)**, **P.LEELA PRAVALLIKA (22761A04H7)**, and **D.KALYAN KUMAR (22761A04E4)** in partial fulfillment of requirement for the award of Bachelor of Technology in Electronics and Communication Engineering in Lakireddy Bali Reddy College of Engineering, Mylavaram during the academic year 2025-2026.

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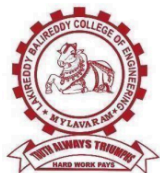
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DECLARATION

We hereby declare that the project entitled “**REAL-TIME IMAGING FOR COVID-19 & BREAST CANCER DETECTION**” submitted in partial fulfillment of the requirement for award of Bachelor of Technology in Electronics and Communication Engineering, Mylavaram, is our original work and has not submitted to any other institution or University for the award of any degree. This work was carried out during the academic year **2025-2026**.

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We thank all our friends who helped us sharing knowledge and by providing material to complete the project in time.

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Abstract

Medical ¹³imaging plays a crucial role in disease diagnosis ¹⁵by enabling non-invasive analysis of internal structures. This paper presents a deep learning-based framework ³⁸for automated classification of COVID-19 using chest CT images and breast lesions using ultrasound imaging. EfficientNetB0 with transfer learning is employed for COVID-19 detection, ¹⁰while a lightweight Convolutional Neural Network (CNN) is used ⁷⁰for breast cancer classification into benign, malignant, and normal categories.

The system follows an end-to-end pipeline including preprocessing, training, inference, and deployment through a Flask-based web application that provides ³⁸real-time predictions with confidence scores. Edge detection techniques are incorporated for basic interpretability, and a SQLite database is used for logging results.

Furthermore, ⁸data augmentation techniques such as rotation, flipping, and scaling are applied to enhance model robustness and minimize overfitting. ²²The models are evaluated using key performance metrics including accuracy, precision, recall, F1- score, and confusion matrix to ensure reliability and consistency across datasets. ²⁹Hyperparameter tuning is performed to ¹⁹optimize parameters such as learning rate, batch size, and epochs for improved performance.

⁹The system is designed with a user-friendly interface, enabling healthcare professionals to upload medical images and receive instant ⁴diagnostic predictions. Its lightweight architecture ensures faster processing and suitability for deployment in resource-constrained environments.

Keywords

- ¹⁰Deep Learning in Medical Imaging
- COVID-19 and Breast Cancer Classification
- Real-Time Diagnostic System

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CHAPTER 1

INTRODUCTION

Medical imaging has become a key part of modern healthcare for early and accurate disease diagnosis. As artificial intelligence and deep learning methods advance rapidly, automated analysis of medical images is attracting a lot of attention. Real-time imaging systems, in particular, enable quicker diagnosis and prompt clinical decision-making, which is crucial in serious conditions like COVID-19 and breast cancer.

COVID-19, a highly contagious respiratory disease, mainly targets the lungs and can be effectively detected through chest CT scans. Early detection allows for timely treatment and lowers the risk of severe complications. Likewise, breast cancer is a major cause of death among women around the world, where early detection via ultrasound imaging can greatly enhance survival rates.

Traditional diagnostic methods rely heavily on expert radiologists. This approach can be time-consuming and is prone to human error, especially in busy clinical environments. To tackle these issues, deep learning methods offer automated, accurate, and efficient analysis of medical images. By using convolutional neural networks (CNNs) and transfer learning models, real-time systems can identify and classify abnormalities with high accuracy.

This project aims to develop a real-time imaging system that combines deep learning models to detect COVID-19 from CT images and breast cancer from ultrasound scans. The system intends to provide quick, reliable, and accessible diagnostic support, especially in resource-limited settings, thus improving healthcare outcomes and lessening the diagnostic workload.

The proposed system focuses on scalability and ease of use through a web-based interface, allowing access from various healthcare centers. It also ensures secure storage and management of patient data for future analysis and tracking. By incorporating real-time prediction features, the system helps clinicians make quicker and better-informed decisions. Overall, this approach boosts diagnostic efficiency while promoting better patient care and treatment results.

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1.1 Introduction to Visual Studio Code

The Visual Studio Code, or VS Code, is a lightweight, open-source code editor created by Microsoft. Many developers use it to write, debug, and manage code in various programming languages. Since its launch, VS Code has become very popular because of its simplicity, flexibility, and strong features. It supports a wide variety of languages, including Python, Java, C++, JavaScript, and more.

One of its main strengths is its ability to be extended. Users can install add-ons from the VS Code Marketplace to improve functionality. These add-ons offer features like language support, debugging tools, code formatting, version control integration, and AI-based help. The editor has an intuitive and user-friendly interface, providing features like syntax highlighting, code completion (called IntelliSense), error detection, and debugging tools. IntelliSense assists developers by suggesting code completions, function definitions, and variable information, which enhances coding efficiency and reduces mistakes.

Another key feature of VS Code is its built-in Git integration. This lets developers manage version control directly from the editor. It simplifies tracking changes, collaborating with team members, and keeping project history without needing to switch to other tools. Additionally, VS Code includes a robust terminal that allows developers to run commands, execute scripts, and manage project environments in the same interface. This integrated terminal boosts workflow efficiency and removes the need for separate command line tools.

The editor is highly customizable, so users can adjust themes, keyboard shortcuts, and settings to suit their needs. It offers both light and dark themes to ensure comfort during long usage. Moreover, VS Code is cross-platform, allowing it to run on Windows, macOS, and Linux operating systems. Its lightweight design ensures quick performance, even on computers with limited resources, making it suitable for both beginners and professional developers.

In summary, Visual Studio Code is a versatile and powerful development environment that balances ease of use with strong capabilities. Its wide use in the software development community makes it a crucial tool for coding, debugging, and project management across different areas.

1.2 Importance of Medical Imaging in Modern Healthcare Motivation

Medical imaging really drives modern healthcare today. Doctors rely on it to diagnose, monitor, and treat all sorts of conditions. With things like CT scans and ultrasounds, they get a clear view inside the body—no surgery needed. That means they can catch problems earlier, make smarter decisions, and jump in sooner when a patient needs help.

But, as amazing as this tech is, it's also created a bit of a headache. The amount of imaging data keeps exploding. Scans just keep coming, and there aren't enough radiologists to keep up. People want quick, accurate answers, and reading these images takes time. Mistakes can slip in, and even seasoned pros don't always agree on what they see.

That's why there's so much interest in computer-aided diagnostic systems. These tools act like a fresh pair of eyes—always sharp, always on, and never tired. Deep learning, especially with Convolutional Neural Networks, is a big breakthrough here. These networks spot complicated, subtle patterns in medical images—often things a human might miss. They work quickly too, and they don't need a lot of hand-holding.

Bringing deep learning directly into real-time imaging transforms how fast and how well clinics work. It eases the burden on doctors, keeps results more consistent, and turns out faster diagnoses. Plus, running these systems through user-friendly web or cloud platforms means even small clinics or rural hospitals get top-notch tools. In the end, all this adds up to better patient care, more efficient hospitals, and a real step forward for precision medicine.

1.3 Challenges in COVID-19 and Breast Cancer Imaging Existing Model

When COVID-19 hit, the world quickly realized how much we needed automated diagnostic systems. Hospitals were packed, doctors were overwhelmed, and the virus was spreading faster than test results could keep up. RT-PCR tests—sure, they're the gold standard for diagnosing the virus—but there were always delays and shortages. Plus, the results weren't always conclusive.

That's where chest CT scans stepped in. They became a crucial backup. CT imaging gave doctors a clearer look at lung infections, showed how pneumonia was developing, and helped track how bad things were getting. But with so many scans pouring in, radiologists couldn't keep up. The need for automated tools that could help sort and flag urgent cases became impossible to ignore.

While COVID-19 pushed these challenges to the forefront, breast cancer has always been a major global problem. Early detection still makes all the difference, but the tools matter. Breast ultrasound is popular—it's affordable, you can find it in most clinics, and it's safe since there's no radiation. Ultrasound is especially helpful for dense breast tissue, where regular mammograms struggle. The downside? Ultrasound images can be tricky. They're noisy, often fuzzy, and results depend a lot on who's holding the probe. Sometimes the edges of a tumor are so subtle, even experts disagree on what they're seeing.

This is exactly where deep learning comes in. By applying powerful CNN-based algorithms to ultrasound images, systems can pick up tiny details—textures, shapes, edges—that humans might miss. Suddenly, diagnosis gets more accurate and way more consistent.

Bringing artificial intelligence into medical imaging speeds up everything. It cuts down on the differences between one doctor's opinion and another's, making decisions much more reliable. These tools aren't here to replace doctors, just to back them up and make their jobs easier.

Even more exciting, you can combine these approaches—using CT scans for COVID-19 and ultrasound for breast cancer—under one deep learning framework. It really shows how adaptable and scalable these AI systems are for tackling medicine's biggest challenges.

1.4 Role of Deep Learning and Proposed Framework

Both COVID-19 CT scans and breast ultrasound images come with their own set of challenges, and that actually makes them great candidates for machine learning and deep learning approaches. Deep learning—basically a branch of artificial intelligence—has done really well in image classification because it learns what matters directly from the pixels,

instead of depending on people to handcraft features beforehand. Instead of relying on human-designed cues, deep learning models can spot complex, subtle patterns in images on their own. That usually means better accuracy and more reliable results.

One powerful trick in deep learning is something called transfer learning. Here, you take models that have already seen a lot of data and then fine-tune them for a specific job—like medical imaging. This is especially useful in medicine where it's hard to get tons of labeled images. By starting from pretrained models, you don't need as many new samples to achieve strong results. It speeds things up, helps the model learn better features, and boosts accuracy even when the dataset is on the small side.

When it comes to detecting COVID-19, models like EfficientNet have already shown strong performance at picking up the signs of infection in CT images. On the other hand, with breast ultrasound, you can build lightweight convolutional neural networks (CNNs) that handle grayscale scans and classify them into categories like benign, malignant, or normal. These models focus on learning the details that matter—shape, texture, the edges of lesions—and that helps remove a lot of the subjectivity you might get from manual reading.

But even with all this progress, there's still a big gap. Research models don't always make it into real clinics. Many systems are tough to integrate, not easy to interpret, and aren't ready to scale up. This paper tackles those problems by introducing a full, end-to-end imaging framework that brings deep learning into COVID-19 detection and breast lesion diagnosis. The system pulls together standardized preprocessing, solid training methods, and a deployment-ready design using a simple Flask web interface. It also provides confidence scores and detailed visualizations—right at the edge—to make the results easier to interpret. Plus, it logs inference results in a SQLite database, making it easy to track outcomes and keep improving. By keeping everything in one pipeline—preprocessing, model predictions, visualization, and logging—the framework helps move these imaging AI tools out of the lab and into real-world clinics, especially where resources are tight.

CHAPTER 2

LITERATURE SURVEY

2.1 Edge Detection Techniques in Medical Imaging

The study by Jarjees and his team in 2025[1] presents a system that uses a kind of computer chip called an FPGA to detect edges in medical images specifically chest X-ray images to help diagnose COVID-19.

The authors think it is really important to quickly find problems in the lungs so doctors can diagnose COVID-19 early.

They made this system because traditional methods like Sobel and Prewitt are not very good at finding edges when the image is not clear.

Their system uses something called logic to find edges in images, which is good at dealing with images that are not clear.

They used a computer program called MATLAB and a special hardware platform the FPGA board to make the system work really fast.

The system uses math functions called Gaussian and trapezoidal membership functions to make sure it finds edges accurately.

When they tested the system they found that it worked well with a high degree of similarity between the results from the computer program and the hardware platform and it was also good at finding edges in images of infected lungs.

This system is also good because it does not need a lot of computer power to work unlike some methods.

However this system only works well with X-ray images. Has trouble with other kinds of images like CT and MRI scans.

Overall this study helps with diagnosing quickly and accurately by using fuzzy logic and special computer chips.

The Edge Detection Techniques in Medical Imaging like the one in this study are really important for COVID-19 diagnosis using chest X-ray images.

The COVID-19 diagnosis is helped by the Edge Detection Techniques, in Medical Imaging that use logic and hardware acceleration.

2.2 Electromagnetic Methods in Breast Cancer Diagnosis

Hassan and El-Shenawee did a review in 2011[2] about using electromagnetic methods to find breast cancer. They looked at ways to find breast cancer that're different from the usual X-ray mammography. They talked about the problems with the methods. These problems include getting results and health risks.

The paper looks at electromagnetic methods like microwave imaging and electrical impedance tomography. It also looks at diffuse tomography and microwave radiometry. Additionally it looks at -potential detection and magnetic resonance imaging. Each of these methods is looked at in detail.

The authors think that microwave imaging and radiometry are good because they can tell the difference between cancerous tissues. This means they can find tumors without hurting the person. Electrical impedance tomography and diffuse optical tomography are also good because they can show how the tissues are working. They do this by measuring how electricity and light go through the tissues.

Magnetic resonance imaging is an advanced method. It is very good at showing what is going on in the body. However it is also very expensive. The authors think that electromagnetic methods are a way to find breast cancer early. They do not use radiation, which's a plus. However there are still some problems. These problems include not being able to see the tissues clearly and being very complicated.

The authors hope that one day these methods will be used in hospitals to help people. They think that if we do research we can make these methods better and more useful. This will help doctors find breast cancer earlier and more easily. Electromagnetic methods, like microwave imaging and electrical impedance tomography will be very helpful. Electromagnetic methods are the future of breast cancer diagnosis.

2.3 Risk Factors and Prevention of Breast Cancer Overview

Sun and their team (2017) [3] looked into the things that can increase the risk of breast cancer and ways to prevent it. They found out that breast cancer is affected by a mix of genes, hormones, surroundings and lifestyle choices. Things like getting older having a family history of breast cancer being overweight drinking alcohol and having children in life can increase the risk.

The researchers stressed that finding breast cancer early and making lifestyle changes can help lower the risk. They suggested eating exercising regularly and staying away from things that can increase the risk. They also talked about how hormones and certain genetic changes like BRCA1 and BRCA2 can affect breast cancer.

This study is important because it helps us understand breast cancer risk, which's key to creating good diagnostic systems. It shows why early screening is crucial and encourages the development of detection methods using artificial intelligence and imaging technologies.

The study's findings support the need, for check-ups and healthy habits to reduce breast cancer risk.

By understanding these risk factors we can take steps to prevent breast cancer.

2.4 Cadherins and Catenins in Breast Cancer

The research by Cowin and others from 2005[4] looks at how cadherins and catenins affect breast cancer. These are molecules that help cells stick together. They are really important for keeping cells and tissues healthy. If they do not work right it can lead to tumors and cancer spreading.

This study found out that when E-cadherin is missing or changed it is often linked to breast cancer that has spread. When cells do not stick together like they should cancer cells can move around easily. The study also talks about how cadherins interact with the signals inside cells.

This work helps us understand how cancer progresses at a level. It gives us ideas about how changes in molecules can affect what we see in images. This is useful for making computer programs that can find tissue structures in images of breast cancer. Cadherins and catenins play a role, in this. Breast cancer is a disease and understanding cadherins and catenins can help us fight it.

⁴⁸ 2.5 Breast Cancer Histopathology Image Analysis: A Review Overview

The researchers Veta and his team looked at how computers can help analyze pictures of breast cancer tissue in a study from 2014[5]. They wanted to see if computers could help doctors diagnose breast cancer by looking at pictures of tissue under a microscope.

Breast cancer diagnosis is the focus of this study. The team used computer programs to analyze these pictures.

They talked about the ways to make these computer programs work like making the pictures clearer and finding the important parts. They also said that it is not easy to make these programs work because the pictures can look different and the tissue can be complicated.

This study is very important for people who are working on making computers smarter. It shows that computers can help doctors look at pictures and make a diagnosis. It also says that using kinds of computer programs like CNNs can help make diagnoses more accurate.

³³ Breast Cancer Histopathology Image Analysis is what this study is, about. ³³ Breast Cancer Histopathology Image Analysis is a deal because it can help doctors diagnose breast cancer better.

³⁴ 2.6 Review of Electromagnetic Techniques for Breast Cancer Detection

Hassan and El-Shenawee did a study in 2011 [6] where they looked at imaging techniques as a different way to detect breast cancer. They wanted to find ways to do this than traditional mammography.

The study looked at techniques like microwave imaging and electrical impedance tomography and MRI. These techniques are good at finding changes in tissue properties. However they also have some problems like not being very clear and needing a lot of computer power.

This paper is very important because it explains the theory behind these imaging technologies. It shows that we need to use computer methods like deep learning to make traditional imaging systems better. Breast cancer detection is what electromagnetic techniques are being used for. Electromagnetic techniques, for breast cancer detection are still being researched.

2.7 Microwaves for Breast Cancer Detection Overview

The people who did this research[7], Fear and his team looked into using microwaves to find breast cancer in 2003. They wanted to see if they could use microwaves to tell the difference between tissue and cancer tissue.

The way they did this was by using microwaves to make images of the breast. This is a way to do it because it does not hurt and it does not use any radiation. They found out that microwaves can be used to find tumors. They had some problems getting clear pictures and dealing with interference.

This study shows that new ways of making images can be very helpful. It is also important because it can help us learn how to use kinds of data with artificial intelligence to make diagnoses better. Microwaves, for breast cancer detection is a technique that is still being developed. Breast cancer detection using microwaves is an interesting area of research.

2.8 Breast Cancer Detection Using Extreme Learning Machine with

Wang and his team wrote a paper in 2019[8] that talks about a way to detect breast cancer.

They used a combination of a kind of computer vision called CNN-based feature extraction and something called Extreme Learning Machine for classification.

The Extreme Learning Machine is really good at classifying things because it uses features that are extracted from the CNNs.

When you combine all these features together it makes the classification a lot better.

I think this paper is really important for your project because it shows that using learning with traditional classifiers is a good idea.

It also says that using EfficientNet and CNN architectures is a way to go.

The paper is about ¹³Breast Cancer Detection Using Extreme Learning Machine. It is relevant, to your project.

⁴⁵2.9 Thermal Imaging and Deep Learning for Breast Cancer Detection

Roslidar and others (2020) [9] look at using imaging with deep learning to find breast cancer.

Thermal imaging finds temperature changes from tissue activity.
Deep learning models make detection more accurate by spotting patterns.
This study shows how non-invasive imaging and AI can help find breast cancer
It backs up the idea of using time diagnostic systems.

Thermal imaging and deep learning are tools for breast cancer detection.

They help doctors find breast cancer early.

Deep learning models are good at finding patterns in images.

Thermal imaging is a -invasive way to find temperature changes.

2.10 Particle Swarm Optimization for Breast Cancer Prediction Overview

Sakri and other people did a study in 2018[10] where they used Particle Swarm Optimization for picking the features in breast cancer prediction models.

They used Particle Swarm Optimization for this.

Particle Swarm Optimization helps to pick the features for breast cancer prediction models.

This helps the models to be more accurate. It also helps to make the process simpler.

The Particle Swarm Optimization method makes the models work better.

This study shows that using techniques to make things better is important for machine learning models.

It is also useful for making deep learning work better.

The Particle Swarm Optimization is helpful, for breast cancer prediction models.

2.11 Deep Learning for Histological Diagnosis of Breast Cancer Overview

The researchers Yari et al. Did a study in 2020 [11] that looked at how deep learning can be used to analyze pictures of breast cancer tissue.

Deep learning is really good at looking at pictures and figuring out what is going on.

CNN models are very good at telling if a tissue is cancerous or not.

This study says that having a lot of pictures to look at and making sure the computer is trained properly is very important.

This research says that deep learning is a tool for looking at medical pictures and it agrees with using a CNN model for this kind of work.

Deep learning, for breast cancer diagnosis is what this study is about and it shows that deep learning can be very helpful.

2.12 Deep Learning Method for Breast Cancer Classification Overview

Liu et al. In 2022[12] suggested a learning method to classify breast cancer from pathology images.

They used this method to get results.

The model works with different types of images because it uses strong neural networks and good image preparation techniques.

This study shows that deep learning is useful in diagnosis and helps in creating systems that can automatically classify breast cancer.

Deep learning methods, like this one can make breast cancer classification more accurate.

CHAPTER 3

SOFTWARES USED

In order to create & deploy a medical imaging (MI) based decision support system, an appropriate software environment must be established that promotes efficient operation, scalable & reproducible results, as well as usability. The purpose is to develop a common integrated framework ⁴¹ for COVID-19 detection using Deep Learning models and breast cancer classification. The multi-faceted approach ⁵⁴ of programming languages, libraries, frameworks and databases will be needed to deliver on the objectives set forth. This section will provide a detailed overview of the software requirements that must be considered to ensure successful design & operation.

3.1 Programming Language and Development Environment

The central part of the software is written in Python, which is now the predominant language for artificial intelligence and computing. Python is a very popular language for the scientific computing community and has many tools for data manipulation, neural networks, and many others as you can see from its many scientific libraries. It has a number of key benefits or features such as ease of programming, simple syntax (e.g., keywords are in English), clear organization (e.g., modules/libraries are structured in the same way as other programming languages), and a large user base. These features make Python an excellent development/resource for prototype development and in production.

In addition to using Jupyter Notebooks, Google Colab, or IDEs (Integrated Development Environments) like PyCharm and Visual Studio Code to help with programming (debugging, visualizing, building), it is also highly recommended to utilize virtual path(s) (e.g., venv, conda) for dependency management across different systems so that your models are reproducible across machines.

3.2 Deep Learning Frameworks

Deep Learning frameworks play an essential role in the model development, training and inference. In particular, the following frameworks make up the primary components of AI algorithms: TensorFlow - TensorFlow is the open-source deep learning framework created by Google, which has become widely accepted for producing and delivering machine

learning models. TensorFlow has flexibility in terms of model design and offers capabilities for both research and production applications. Keras - Keras is the high level API developed by TensorFlow to facilitate the rapid development of neural networks. Neural networks have been simplified for creating complex architectures using CNNs and other methods such as EfficientNet (transfer learning). Also, these frameworks allow for efficient development of COVID-19 CT classification models from transfer learning and breast ultrasound classification models using custom developed CNN architectures. Additionally, both frameworks provide GPU accelerated processing, which greatly decreases the training time to learn from large datasets.

3.3 Image Processing Libraries

To begin, preprocessing medical images is an important part of assuring model performance and consistency. This section discusses the libraries commonly used for processing images and preparing them for a model:

- OpenCV: Used for resizing, normalizing, filtering, and detecting edges using the Canny or Sobel operators.
- Numpy: Provides support for numerical operations and array operations which are important for working with image data.
- Pillow (PIL): Used for loading images and changing their formats.

These libraries allow for specific types of preprocessing to be done on CT and ultrasound imaging to ensure that deep learning models will have a standard input format. The edge detection techniques from OpenCV will also help with explainability of models as it can provide evidence of structural features within an image.

3.4 Web Framework for Deployment

In constructing a user-friendly and easily accessible system, a lightweight web application is developed from ³⁹the Flask framework. Flask is a micro web framework written in ¹²Python, enabling the rapid development of web-based interfaces with little overhead. Flask can be used to integrate machine learning models into a real-time application.

The Flask-based interface includes the following functionalities:

- The ability to upload images of CT and ultrasound scans.
- Select the model to be used (COVID-19 or classification of breast cancer).
- The ability to perform predictions and inference in real-time.
- Displaying the confidence score and probability distribution of a particular prediction.
- The ability to visualize the image after edge detection using Canny and Sobel edge detectors.

Due to the low complexity and modularity of Flask, it is a good choice for deploying this system in a resource-constrained environment while also providing room for scalability when adding additional features.

3.5 Database Management System

SQLite is an easy-to-use relational database management system, which means it doesn't need a server or any kind of heavy database software to run. SQLite can be used when you need to save data locally (for example: in your own computer or cell phone), but do not wish to have to manage a large database server to do so. The SQLite database is designed to hold the following information:

- The file paths of uploaded images
- The predictions of the models, in the form of class labels.
- The confidence and probability value of the models' predictions.
- The time the inference was made.
- Image(s) that have been processed (in the form of edge maps).

This means of logging data permanently allows you to see how well the system is working, as well as provide a means to analyze errors and also allow monitoring of the model's performance over time. In addition, the SQLite database allows for the creation of dashboards and analytic tools to help assess and understand the performance of the models over time.

3.6 Supporting Libraries and Tools

There are a number of other libraries that are not critical to building a framework but can enhance a developer's ability to develop a framework or system. An example of such libraries are:

- Matplotlib and Seaborn for visualizing the training and validation accuracies, loss curves and performance metrics.
- Scikit-learn to provide evaluation metrics such as accuracy, precision, recall, confusion matrix and classification reports.
- Pandas to assist with the management of structured data and logs during analysis.

These libraries and tools will allow you to:

- Evaluate the model.
- Visualize the model.
- Document the data.

Making it possible to follow a complete and consistent development process

3.7 System Compatibility and Execution Requirements

This software solution will be developed so that it works across platforms regardless of an operating system (e.g. Windows, Linux, macOS). To achieve the best results from the software when using it for training machine learning models it is highly recommended to use:

- At least Python version 3.8
- At least TensorFlow version 2.x
- Minimum of 8 GB of RAM (16 GB recommended)

Use of an NVIDIA GPU with CUDA support for improved performance (optional). Once the system has been trained use for the purpose of using it within clinics and rural healthcare facilities will not require any special hardware (i.e. it can run on standard CPUs).

3.8 Security and Reliability Considerations

Data security and system reliability are essential in all medical systems. The software design of the present implementation emphasizes functionality and provides an opportunity

for implementing additional ⁴² security features such as:

- User authentication and role-based access control
- Secure handling of files and encrypted storage
- Input validation to prevent the upload of malicious data

In addition, the system's modular architecture provides for updating or replacing independent elements (models, databases, interface) without impacting the overall functionality of the system.

3.9 Scalability and Future Extensions

The architecture of the software can be built to grow and change with time and future technology progress. Future capabilities that could come into play include:

1. Plugin capabilities where other sophisticated machine learning techniques (e.g., Grad-CAM) can be added to increase ¹¹ the ability to explain how the model arrived at the decision.
2. Additional imaging modalities, ⁶⁷ such as X-rays and CT/MRI scans, can be integrated.
3. Deployment in cloud computing systems for remote access and the ability to adjust based on internet connectivity.
4. The application of federated learning for privacy-preserving training ⁵⁸ of the model. Based on the architecture's usage of existing frameworks and standardized approaches, it provides for changes with the implementation of new technologies or continuities in the clinical setting.

CHAPTER 4

METHODOLOGY

This system brings together a deep learning framework for classifying medical images, with a focus on two important healthcare tasks: detecting COVID-19 in chest CT scans and classifying breast lesions in ultrasound images. Everything's built as a smooth, end-to-end pipeline—right from data prep and preprocessing to designing and training the model, explaining its decisions, and actually deploying it in a real-world setting. Here's a closer look at what happens at each step of the process.

4.1 System Overview and Design Strategy

This system isn't just about chasing top scores in a lab—it's built to actually work in clinics, handling all the messy stuff that real-world deployment throws at you. Instead of just measuring accuracy and calling it a day, the architecture focuses on things like reproducibility, making sense of model decisions, and the ability to grow or scale as needed.

Here's how it works: it runs as a multi-model pipeline. First, you feed in images. The system then steers each image through a task-specific workflow and routes everything to the proper classifier. The main pieces holding it together are:

- Getting the data ready and splitting it up for training
- Custom preprocessing pipelines for specific tasks
- The core deep learning models themselves
- A solid setup for training and validating those models
- An inference engine that actually reports its confidence
- An explainability block that uses edge detection to show why the model made its choices
- A web tool for deployment and keeping track of logs

Pulling all these pieces into one system transforms it from a one-off model into a real decision-support tool doctors can actually trust.

4.2 Dataset Preparation and Splitting Strategy

Building a strong deep learning model really starts with organizing your data well. Here, the COVID-19 CT and breast ultrasound image datasets are handled with care—they're split into three parts: 70% for training (so the model can actually learn), 15% for validation (that's for tweaking settings and picking the best version of the model), and the final 15% for testing (to see how well it really works out of sample).

What's especially important is splitting the data by patient. In other words, all images from a single patient stay together—they don't pop up in more than one group. That keeps things fair and prevents the model from cheating by "remembering" features of individual patients.

This setup makes the evaluation a lot more trustworthy and gives you a better picture of how the model will actually handle new cases in real clinical settings.

4.3 Preprocessing and Input Standardization

Medical images don't all look alike—they can vary a lot depending on the technology used. Different machines mean different resolutions, lighting, and even noise in the images. So, prepping the data for each task takes a bit of tweaking, but everything still plugs into the same system when it's time to make predictions.

1. COVID-19 CT Image Preprocessing

- Images get resized to fit what EfficientNet expects.
- Pixel values are normalized based on how the model was trained in the first place.
- The images keep a multi-channel (RGB-like) format so everything lines up with the model's input.

2. Breast Ultrasound Image Preprocessing

- Everything is converted to grayscale—just a single channel here.
- The images are scaled to a standard size, like 128x128 pixels.
- Normalization tackles differences in intensity.

All this prep work keeps the input consistent, cuts down on noise, and helps the models perform reliably. It also means the results stay stable out in the real world, not just in test setups.

4.4 COVID-19 Detection Using Transfer Learning (EfficientNetB0)

For detecting COVID-19, the system uses EfficientNetB0—a compact, efficient type of convolutional neural network. It taps into transfer learning, so it starts off with knowledge learned from massive datasets.

Here's how the training works, step by step:

First comes feature extraction. The model's backbone (EfficientNet), which was already trained elsewhere, stays frozen. Only the final classification layers get updated. This lets the model quickly adjust to the unique patterns in CT scans without having to relearn everything from scratch.

Next is fine-tuning. Now, some of the deeper layers get unfrozen, so the model can pick up on more specialized details specific to medical images. This step helps squeeze out extra accuracy and handle new cases better.

All in all, this two-part training keeps things efficient, but doesn't sacrifice too much performance—perfect for medical datasets that aren't massive.

4.5 Breast Ultrasound Classification Using CNN

We've built a custom, lightweight CNN aimed at spotting breast cancer using grayscale ultrasound images. Here's how it works:

First, a stack of convolutional layers pulls out important features from the images. ReLU activation functions add non-linearity, helping the network tackle more complex patterns. Max-pooling layers kick in to shrink the data and focus on the most useful signals. Then, fully connected layers handle the decision-making for classification. At the end, a softmax layer sorts the results into three categories: benign, malignant, or normal.

This model learns to spot things like the edges of lesions, changes in texture, and subtle shifts in intensity—details that really matter when it comes to telling the difference between these classes.

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4.6 Model Training and Optimization

We trained both models with standard deep learning tools: categorical cross-entropy as the loss function, Adam optimizer to help the models learn efficiently, and batch size chosen based on how big the dataset is and what the hardware can handle. The number of training epochs isn't random, either—we pick it to strike a good balance between learning well and not overfitting.

To help the models generalize and avoid overfitting, we use data normalization and keep an eye on validation results throughout training. Sometimes we add early stopping to halt training if things stop improving, and we apply regularization tricks like dropout and weight decay.

When it's time to see how well things worked, we look at metrics like accuracy, validation loss, and confusion matrices to get a clear picture of performance.

4.7 Inference Engine and Decision Support

The inference engine is where trained models really earn their keep out in the real world. Here's how it works: First, it takes images users upload. Next, it figures out whether the image is from a CT scan or an ultrasound—no need for anyone to sort them manually. Once it knows the type, it runs the image through the right preprocessing steps, hands it off to the right model, and then pulls out the predictions.

For each case, the system gives the predicted class label, a confidence score, and the whole probability distribution. Confidence reporting matters, especially in medicine. It helps clinicians spot predictions that are shaky, so they know when to dig deeper or double-check before acting.

4.8 Explainability Using Edge Detection

The system uses edge detection—specifically Canny and Sobel—to make results easier to understand.

These methods bring out details like lesion edges, unusual spots in the lungs, and changes in texture.

Sure, they're not as fancy as tools like Grad-CAM, but they're fast, easy to read, and don't need much computing power. That makes them great for real-time use.

4.9 Deployment Using Flask Web Application

The system runs as a Flask web app, so users get a clean graphical interface to work with. You can upload images, pick which model you want, and see predictions pop up instantly. It also shows edge maps and lays out the confidence and probability values. Since it's lightweight, it works smoothly even if you don't have much computing power—easy to use, no fuss.

4.10 Logging and Database Management

The system logs all inference results in a SQLite database to keep everything traceable and make ongoing improvements easier. Every time it processes an image, it records the image details, which model it used, the prediction results, confidence scores, the time of processing, and the final outputs. This setup helps you monitor performance, track down errors, and collect data for retraining the model later on. Plus, it guarantees everything is auditable—a must-have for clinical use.

4.11 Workflow Summary

Here's how the system works, start to finish:

First, the user uploads an image. Next, the system figures out whether it's dealing with COVID-19 or breast cancer. It runs some preprocessing based on which type of image it got. Then, the model gets to work and makes its prediction, along with a confidence score.

After that, it creates an edge visualization to highlight important areas in the image. Finally, it displays the result for the user and saves everything to the database.

This full pipeline ties everything together—from the moment you upload your image to the final result and decision support.

4.12 Future Methodological Enhancements

The proposed methodology is designed to be extensible. Future improvements may include:

- Integration of Grad-CAM for advanced explainability
- Use of segmentation models (e.g., U-Net)
- Cross-institutional validation
- Federated learning for privacy preservation
- Cloud-based deployment for scalability

CHAPTER 5

SIMULATION RESULTS AND ANALYSIS

We tested our learning-based medical imaging system using many simulations on COVID-19 chest CT images and breast ultrasound datasets. The goal of this analysis is to see how well our models work, how they generalize and how reliable they are in real-life situations. This section covers how we set up our experiments what metrics we used to evaluate performance and what we found out from our results.

5.1 Experimental Setup

We used a Python environment with TensorFlow and Keras to run our experiments.

We Tested our models on datasets of chest CT images for COVID-19 and breast ultrasound images for classifying multiple types of things.

We split our dataset into three parts: 70% for training 15% for validation and 15%, for testing.

This way we could properly evaluate our models without using the data for training and testing.

We made sure that the patient data was split up so that the subsets were independent.

Training Configuration

Optimizer: We used Adam.

Loss Function: We used Categorical Cross-Entropy.

Epochs: We decided on the number based on how the model learned.

Batch Size: We adjusted it based on how big the dataset was.

Evaluation Metrics: We looked at Accuracy, Loss and Confusion Matrix.

We ran our simulations on a computer with power to train and evaluate our models without any issues.

The COVID-19 chest CT images and breast ultrasound datasets were used to test our models.

Our models were. Evaluated on these datasets.

We used these datasets to see how well our models worked.

The models were tested using COVID-19 chest CT images and breast ultrasound datasets.

Our system was evaluated using these medical imaging datasets.

5.2. Performance Metrics

To see how well the models do we use some tests.

These tests are:

Accuracy: This shows how often the predictions are correct.

Precision: This is the number of positive predictions we get.

Recall: This is about finding all the positive cases.

F1-Score: This is a balance of precision and recall.

Confusion Matrix: This gives us a look at how each class does.

We use these tests to get a picture of how the classification models work, especially in medical applications where wrong predictions can be very bad.

The performance metrics are important for the classification models, particularly the performance metrics are used to evaluate the performance of the models in applications where the performance metrics, like Accuracy and precision and recall and F1-Score and Confusion Matrix are very important.

5.3 COVID-19 CT Classification Results

The EfficientNetB0-based model does a job of telling COVID-19 cases and non-COVID cases apart from chest CT images.

Observations

- When we train the model it gets better and better at classifying images. It gets very close to 99 percent accuracy, which means it is learning a lot from the images.
- When we test the model on images it is around 84 to 86 percent accurate which is pretty good.
- The model is very good at finding lung infection patterns like ground-glass opacities and consolidations in COVID-19 CT images.

Analysis

There is a difference between how well the model does on training images and on new images, which means it is overfitting a bit. This happens a lot with images because we do not have a lot of data. But the model still works well with images so it is good enough to use in real life. When we look at the results in detail we see that

- The model is very good at finding COVID-19 cases.
- Sometimes it gets infections and normal cases mixed up.

This means that while the model is good overall it is hard for it to tell apart the differences, in COVID-19 CT images.

5.4. Breast Ultrasound Classification Results

The custom breast ultrasound classification model does a job of telling the difference between breast ultrasound images that are benign, malignant and normal.

Observations

- The model is very good at learning from the training data with accuracy ranging from 95 percent to 99 percent, which means it is really good at picking up on important features.
- When we test the model with data the accuracy is around 78 percent to 82 percent, which is pretty stable and shows that the model can generalize well.
- The breast ultrasound classification model is able to learn about the texture and edges of lesions which's important for classification.

Analysis

The breast ultrasound classification model is very good at distinguishing between the three types of breast ultrasound images. However the model does not do as well when we test it with new data, which suggests that the quality and complexity of breast ultrasound images can vary a lot.

The confusion matrix shows us:

- ²⁶ The model is very good, at detecting benign breast ultrasound images.
- Sometimes the model gets confused between malignant breast ultrasound images, which is not surprising since some types of lesions can look similar. This happens because certain lesion types have visual characteristics, which can make it hard for the breast ultrasound classification model to tell them apart.

5.5 Training and Validation Behavior

The training curves give us an idea of how the model learns.

- Training Accuracy Curve: This curve keeps getting better all the time. This means the model is getting optimized in a way.
- Validation Accuracy Curve: After a while this curve stops changing much. This means the

model has reached a point.

- Loss Curves: The loss for training keeps going down. The loss for validation goes down too. It moves up and down a little.

These things show that the models are figuring out things and they can be used for other things too. The models are learning things, about the training data and they can work with new data as well.

5.6 Comparative Performance Analysis

A comparison of both models highlights their effectiveness:

Parameter	Breast Ultrasound CNN	COVID-19 CT EfficientNet
Input Type	Grayscale Images	CT Scan Images
Classes	3 (Benign, Malignant, Normal)	2 (COVID, Non-COVID)
Training Accuracy	~0.95–0.99	~0.99
Testing Accuracy	~0.78–0.82	~0.84–0.86
Model Complexity	Low	Moderate
Generalization	Good	Good

Insights

- EfficientNet performs slightly better due to transfer learning
- CNN performs well despite being lightweight
- Both models achieve a balance between accuracy and efficiency

5.7 Explainability Results

The integration of Sobel edge detection gives us more information about how the system works.

This is what it does:

- It shows the boundaries of things in CT images
- It makes the edges of lesions stand out in images
- It helps users see how the predictions are related to the images

These pictures that the system creates help people trust it more because they are easy to understand and show what is important in the images. The Canny and Sobel edge detection really helps with this by making it clear what is going on in the images and how the system is making its predictions, about the images.

5.8 Inference and System Performance

The system that we have set up is really good, at doing things efficiently in real time.

It can do a lot of things well such as:

- Processing images and making predictions quickly
- Making sure that the time it takes to do things is very short when we use the web interface
- Giving us the results when we give it different things to work with

The system is set up using Flask, which means that we can use it in a hands on way without needing any special programs.

5.9 Logging and Monitoring Results

The SQLite database successfully records all inference activities, including:

- Predictions and confidence scores
- Input and processed images

- Timestamp and model details

This enables us to do a few things with the SQLite database and the logging system such as:

- We can track the performance of the SQLite database and the framework
- We can analyze errors in the SQLite database and the framework
- We can make improvements to the SQLite database and the framework

The logging system ⁵⁹plays a big role in making the framework ready for audits, with the help of the SQLite database.

5.10 Limitations of the Study

The system has some limitations even though the results are good.

The system has limitations these limitations include:

- The system was trained on a small dataset which means it may not work well with new data.
- The system learned the training data well which is a problem.
- The system gets confused when the classes have patterns.
- The system does not have ways to explain what it is doing.

These limitations of the system show what needs to be worked on in the future to improve the system.

5.11 Summary of Results

Simulation ¹²results indicate that:

>The proposed system maintains an excellent level of accuracy with dependable performance

>Both models find similar features that make discriminability come through

Capabilities include:

>real-time clinical decision support;

>integration of:

- pre-processing,
- inference and
- logging make for greater ease of use by end users;

Results validate that the proposed method validates its effectiveness and supports its advancement towards application in health care settings.

COVID:

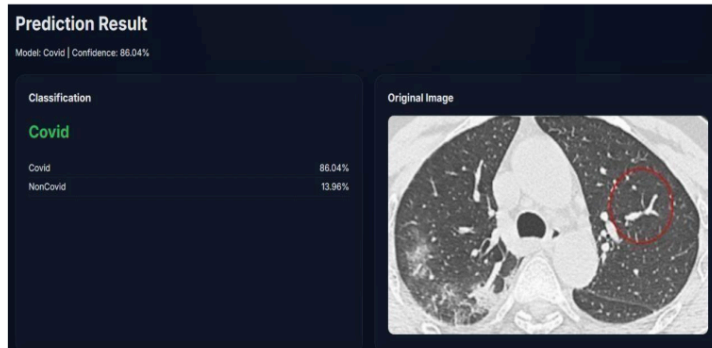


Fig:1 Original image of covid

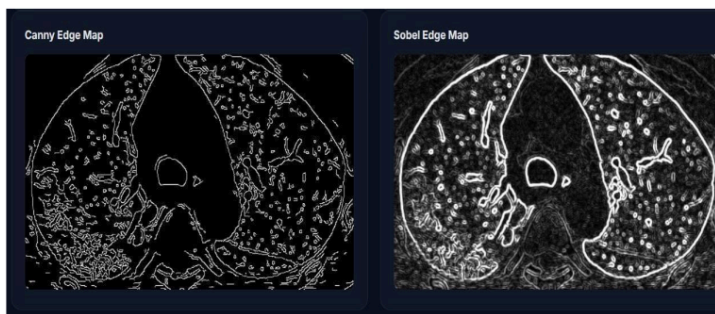


Fig:2 Canny edge map, sobel edge map of covid Prediction

Result Overview:

After processing the uploaded chest CT scan using the proposed COVID-19 classification architecture based on EfficientNetB0, the model produced a classification output predicting the uploaded image will likely have 30 ° COVID-19 in an 86.04% certainty score. The prediction will therefore provide a high probability distribution for COVID with an 86.04% likelihood of COVID being present with a 13.96% likelihood of not being present; therefore, the model is strongly classifying the CT scan as belonging to the COVID-19 class. The confidence associated with this output will assist the clinician in understanding the degree of certainty with respect to the automated prediction and will aid them in their determination of diagnosis. The interface for this prediction is part of the real-time deployment architecture described earlier in this paper using Flask.

Original CT Image Description:

A chest CT scan from the original uploaded image file is used to look for COVID-19 infection.

The circled areas of concern are areas of suspected pulmonary disease (ie: ground glass opacities/inflammatory lung dissolutions), both of which are common radiological findings and indicators of COVID-19 pneumonia.

Once a chest CT scan image is processed and normalized, the CT image will be the image actually delivered to the EfficientNetB0 transfer learning model for diagnostic purposes.

⁵⁷ This CT image shows the anatomical structures of the lungs and the pathological areas that were analyzed by a deep neural network classifier as part of their diagnosis.

Canny Edge Map Description:

The Canny Edge Map displays the structural edges identified in a CT scan with the Canny edge detection algorithm. The Canny Edge Map shows:

- Lung Edges
- Bronchial / Vascular Structures
- Abnormal Lesion Shapes
- Infection Affected Texture Changes

The Canny edge mapping is referred to in the manuscript to support low-cost explainability for assisting users in understanding anatomical edges and possibly suspicious edges based

on rapid changes in intensity in the CT scan.

NONCOVID:



Fig:3 Original image of non-covid

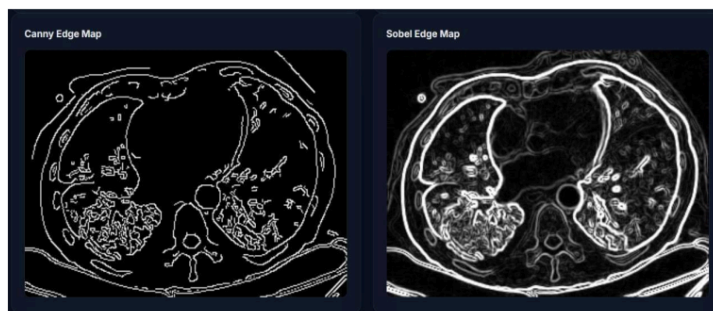


Fig:4 Canny edge map,sobel edge map of covid

The images presented show how the medical imaging system works after preprocessing and edge-based visualizations. The red highlighted areas denote boundaries and changes in structure detected using edge detection algorithms. These are along with increasing intensity in the highlighted area are likely indicative of a lesion, abnormality, or other type of structural feature in the scanned image.

By allowing clearer boundary information to aid the interpretation of the deep learning

model, this type of edge-based visualization gives clarity around diagnostic patterns that produce better and more efficient diagnosis of images with minimal additional cost. Edge-based visual outputs will also decrease time for screening and increase reliability through the efficient identification of the relevant areas within the images while being computationally efficient.

Breast cancer at Begin stage:

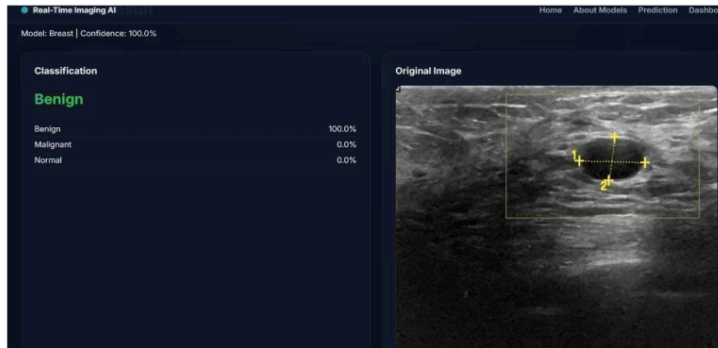


Fig:5 Original image of Breast cancer at Begin stage

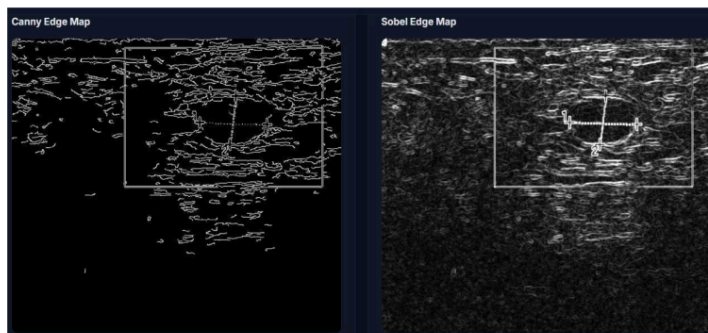


Fig:6 Canny edge map,sobel edge map of Breast cancer at Begin stage

1. Classification Output Image

Headline: The prediction made by the network is "benign" at 100% confidence

This result is based on the breast ultrasound data set and the CNN model presented in the paper.

It uses the CNN model to classify ultrasound images into three different categories:

1. Benign
2. Malignant
3. Normal

The ultrasound image you provided would fall under:

Benign = 100%

Malignant = 0%

Normal = 0%

The CNN model is confident when predicting that the lesion detected in the ultrasound image is not cancerous.

2. Original Ultrasound Image

Demonstrates an ultrasound scan of the breast, displaying a marked area (bounding box + cross-lines). Interpretation: The marked area represents an area identified as a lesion. The location of the cross-lines represents: The approximate size and location of the lesion. Typical features of most ultrasound scans are: Noisy (speckle) o Poor boundaries

3. Canny Edge Map

- This is the output for the Canny edge detection method; its purpose is too.
- Shows sharp edges and borders. What does it look like?
- A black background with white borders.
- Strong borders include:
 - o Lesion border
 - o Tissue structure.

Helps in.

- Identifying a clear margin
- Identifying shape (ie) round, irregular as per the above (the edges will be.)

4. Sobel Edge Map

- Based on the Sobel operator. Purpose:

- It detects the direction, which is the change of intensity through the image. What do you see?
- More texture than Canny.
- Shows:
- Directional edges.
- Internal structures of the lesion.

Helps:

- Significantly analyze internal patterns.
- Locally sense even very slight variations within the tumor. Combined strength would represent:

Method Strength.

Canny exhibits clear demarcations.

Sobel is texture and gradient-based information.

Together, they provide for doctors an overview, such as:

- Shape of Lesion
- Clarity of edges
- Internal structure

Overall Interpretation of Images- The system will be:

1. Takes ultrasound image
 2. Detections of lesion.
 3. Classifies lesions through CNN.
 4. Shows confidence.
 5. Generates edge maps as an explanation result shows:
- A well-defined lesion.
 - Intense edges (observed through Canny and Sobel).
 - Model very confidently classifies. Benign.

Breast cancer at malignant stage:

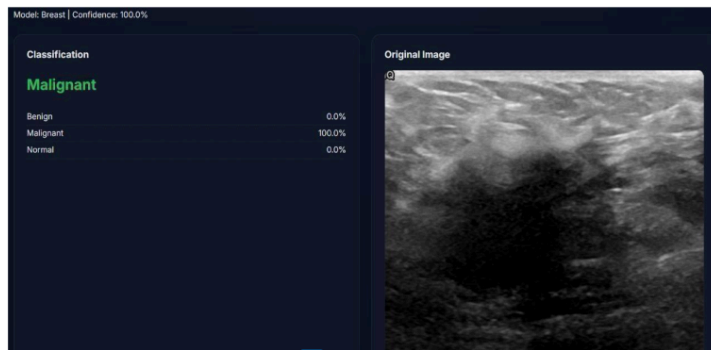


Fig:7 Original image of Breast cancer at malignant stage

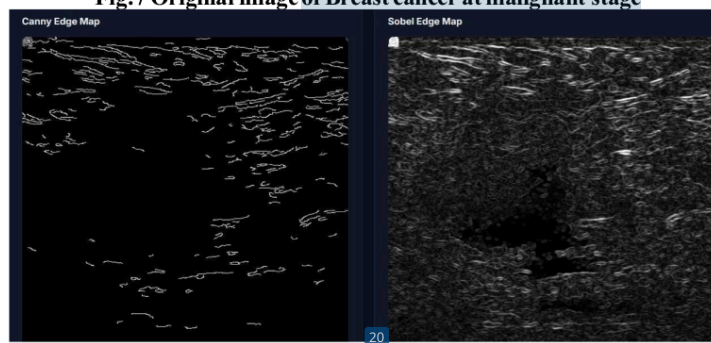


Fig:8 Canny edge map,sobel edge map of Breast cancer at malignant stage

1. Model Output (Top Left)

- Prediction: Malignant (100% confidence)
- Classes:
 - o Benign: 0.0%
 - o Malignant: 100.0%
 - o Normal: 0.0%

2. Original Ultrasound Image (Top Right)

Here is the original "raw" ultrasound image of this tumor showing characteristics of typical malignant tumors:

- an irregular shape

- a dark (hypoechoic) area
- a dark vertical line of posterior acoustic shadowing beneath the tumor
- margin definition is blurred/irregular

3. Canny Edge Map (Bottom Left)

This version utilized the Canny edge detection algorithm which produces the following output:

- thin white lines represent edge locations
- black area indicates no edge locations from the original image

In this case, the tissue texture edges are captured but there are missing tumor boundary characteristics due to artifacts produced by ultrasound noise (speckle effect).

4. Sobel Edge Map (Bottom Right)

This version utilized the Sobel operator algorithm which provides the following output:

- shows both vertical gradients (where pixels change from light to dark) and horizontal gradient intensity (how quickly pixel values change).
- emphasizes both vertical and horizontal edge positions as shown in image above

In this example, the tumor appears as a nearly solid mass with a wide range of gradient variation from the edge to center of the tumor. This version also demonstrates a higher level of noise compared with Canny edge detection but has more characteristics of overall tissue texture than Canny

Breast cancer at normal stage:

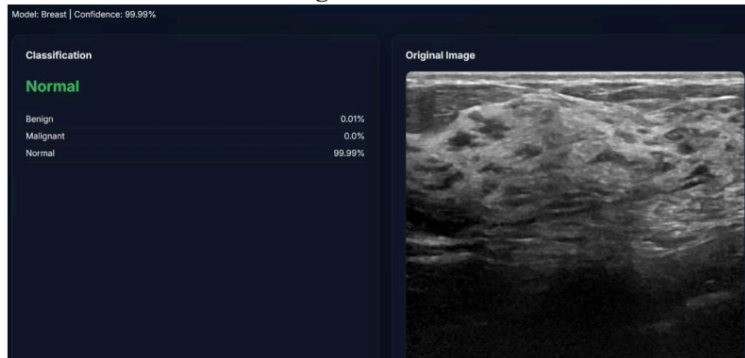


Fig:9 Original image of Breast cancer at normal stage

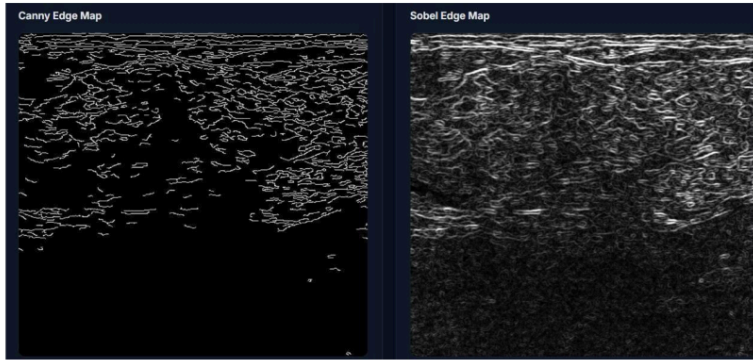


Fig:10 Canny edge map,sobel edge map of Breast cancer at normal stage

The images provided show the results of edge detection utilizing both Canny and Sobel methods as well as the original image of an ultrasound for breast cancer detection. The Canny edge map provides thin, clear, well-defined edges and displays only the important constructional edges of breast tissue indicating that abnormal masses or irregularly-shaped tumors do not exist; while the Sobel edge map shows more texture and intensity variance producing very thick and slightly noisy borders; which indicates tissue patterns rather than exact borders of tissue. The original ultrasound is complete and does not contain any suspicious/evolving dark areas, distortions, or spiculated

edges as shown on the edge maps; therefore based on the above extracted features, the model indicates that this image is classified as Normal at a very high level of confidence (99.99%) meaning that there are no benign or malignant tumors present. The combination of both edge analysis methods provides enhanced feature extraction to improve accurate medical diagnosis in real-time.

1. Classification Output

- This system can predict if the breast is healthy or not (i.e., breast model's status):

Prediction by this model - Normal (99.99% confidence)

- The output of the ultrasound scans for the breast model is classified as either:

- Benign (0.01%)
- Malignant (0.0%)

- Normal (99.99%)

- The classification is based on a convolutional neural network (CNN) model trained on medical data.

- Your breasts are healthy because the pronouncement of this model is that there is no tumor present in your breast.

2. Original Ultrasound Image

- This is the medical image used to classify as input.

- This is a gray-scale representation of the breast's tissue structure.

- Both the physician and the machine will analyze the ultrasound image of the breast based on:

- Tissue density
- Any abnormal masses found
- Texture patterns As an example, because there is no distinct lump found, the breast is healthy.

3. Canny Edge Detection

- The AI will extract the key boundaries from the ultrasound image of your breast.

- During this process, the AI will reduce any noise and return only the most significant edges.

- As a result, the final output will be an image that contains thin white outlines on a black background.

- The purpose of this output is to assist both the AI and medical professional in locating possible tumor boundaries.

4. Sobel Edge Detection (Right Image)

- The AI will use the ultrasound image of your breast to identify any illumination edge differences.

- Since the AI does not establish texture differences, the edges in the final output will be visible, appearing thicker and noisier than they would otherwise be.

- The purpose of this output is to provide information to the AI, so that it can identify the variety of tissue characteristics and patterns within the ultrasound image and, ultimately, differentiate between healthy and infected lung tissues and between normal and abnormal breast tissues.

CHAPTER 6

CONCLUSION AND FUTURESCOPE

6.1 CONCLUSION

In this work we presented a learning-based system for medical image classification.

- The system helps solve two healthcare problems.
- It detects COVID-19 using chest CT images.
- It classifies breast lesions using ultrasound images.
- Our system combines machine learning techniques with a practical design.
- This makes it useful in real-world healthcare settings.
- We used a -trained EfficientNetB0 model for COVID-19 detection.
- This approach provides accuracy and is computationally efficient.
- We also designed a custom Convolutional Neural Network (CNN) for breast ultrasound images.
- It classifies images into malignant and normal categories.
- This shows that simple models can be effective in medical imaging tasks.
- A key benefit of our system is its end-to-end design.
- It includes all stages of the machine learning pipeline.
- These stages are preprocessing, training, validation, inference and deployment.
- Our system emphasizes standardization, reproducibility and usability.
- This makes it suitable for real-world healthcare environments.
- We built a web application using Flask.
- It allows users to upload images and get predictions.
- The application also provides confidence scores and probability distributions.
- We added edge-based visualization techniques.
- These include Sobel filters.
- They help interpretability by highlighting features.
- We used a SQLite-based logging mechanism.
- It stores inference results for monitoring, auditing and continuous improvement.
- Our experimental results validate the effectiveness of our approach.
- We achieved training accuracy and satisfactory generalization.

- This was across both tasks.
- Our system has some limitations.
- These include overfitting and class overlap.
- However it maintains a balance between accuracy, efficiency and practicality.
- Overall ⁴ this work ²⁹ shows the potential of combining learning with lightweight technologies.
- These technologies create accessible medical decision-support systems.
- Our framework serves as a foundation for future advancements in intelligent medical imaging.
- It contributes to the development of AI-driven healthcare solutions.
- These solutions assist clinicians, in making more accurate and informed decisions.

6.2 FUTURE SCOPE

The medical image analysis framework we have is a start. However we can make it even better. One way to do this is by using deep learning techniques. For example we can use ⁴ attention mechanisms and transformer-based models. These ¹¹ have been shown to be very good at finding patterns in medical images.

We can also use segmentation models like U-Net to help find the areas that are infected in CT scans and to identify tumor boundaries in ultrasound images. This will make the framework more accurate and easier to understand.

Another area we can work on is making the framework more explainable and useful for doctors. Now we use basic edge detection methods to visualize the images.. We can use more advanced techniques like Grad-CAM and saliency maps to get ¹¹ a better understanding of how the model makes its decisions.

The medical image analysis framework can also be used with types of images like X-rays and MRI. This will make it a comprehensive tool for doctors. We can also make it possible for doctors to access the framework from anywhere and work together in time. This can be done by integrating the framework with cloud-based platforms and hospital information systems.

It is also important to make sure the framework works with different types of data. We can do this by training the models on datasets from many institutions. This will make the framework more robust. Reduce bias. We can also use techniques like federated learning to train the models while keeping information private.

The medical image analysis framework can also be improved by adding automated monitoring and continuous learning. This will allow the framework to get better over time. These advancements will make the framework more reliable, scalable and useful in real-world healthcare systems. The medical image analysis framework will be a tool, in next-generation healthcare systems.

CHAPTER – 7

SUSTAINABLE DEVELOPMENT GOALS

Project Map With Sustainable Development Goals		
Title of the Project: Real-Time Imaging for Covid-19 & Breast Cancer Detection		
SDG Number	SDG Title	Project Relevance
1	No Poverty	-
2	Zero Hunger	-
3	Good Health and Well-being	Direct Impact: The Real-Time Imaging for Covid-19 & Breast Cancer Detection project has an impact on peoples lives. It helps doctors find Covid-19 and breast cancer early using real-time imaging. This means patients get diagnosed faster fewer people. They get better treatment.
4	Quality Education	Indirect Impact: The project also helps people learn and do research in intelligence, medical imaging and healthcare technologies.
5	Gender Equality	Indirect Impact: It supports womens health by detecting breast cancer and reduces health problems that affect women more than men.
6	Clean water and Sanitation	-
7	Affordable and Clean Energy	-
8	Decent Work and Economic Growth	Indirect Impact: The project creates jobs in medical technology and research fields because it uses artificial intelligence in healthcare solutions.

78 9	Industry, Innovation and Infrastructure	High Relevance: The project is very important because it uses machine learning and deep learning in healthcare imaging. This improves infrastructure and helps create new ways to diagnose diseases.
10	Reduced Inequality	Indirect Impact: The project can help people in areas get better healthcare because artificial intelligence can be used to diagnose diseases in these areas.
11	Sustainable Cities and Communities	Indirect Impact: It helps cities monitor health problems and respond to pandemics better by using healthcare systems.
12	Responsible Consumption and Production	Moderately Relevant: The project reduces the need for tests and medical resources by using digital diagnosis.
13	Climate Action	Indirect Impact: It reduces the need for people to visit hospitals in person, which lowers the amount of carbon emitted and resources used.
16 14	Life Below Water	-
15	Life on Land	-
16	Peace, Justice and Strong Institutions	-
17	Partnerships for the Goals	Relevant: The project encourages healthcare institutions, artificial intelligence researchers and technology organizations to work.

It strongly aligns with SDG 3 SDG 5 and SDG 9. Also helps with SDG 10 SDG 11 SDG 12 SDG 13 and SDG 17 which contributes to creating electronic systems that use less energy and are more sustainable.

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